

UC LINA 600 LINEAR ALGEBRA FOR GRAPHICS · MODULE 02 · LECTURE NOTES

Vectors, Spans, and Bases

What a coordinate actually is.

MODULE OUTCOMES

- Explain a vector as an arrow and as a coordinate list.
- Define span and linear independence.
- Define a basis and explain why coordinates need one.
- Explain dimension.
- Recognise these objects in graphics contexts.

1 Two readings

A VECTOR IS AN ARROW, AND ITS COORDINATES ARE INSTRUCTIONS FOR BUILDING IT FROM BASIS VECTORS

(3, 2) means 'three of the first basis vector, plus two of the second' — so the numbers are relative to a choice somebody made, and the arrow itself is not. The arrow exists; the numbers describe it from a particular point of view.

Which is exactly why the same arrow has different coordinates in different bases — and why change of basis (Module 04) is the central operation of this course rather than a technicality buried in chapter nine.

And the arrow reading is the one to think in, with the coordinate list as bookkeeping — **which is the reverse of how the subject is usually taught, and is why the 3Blue1Brown series is first on the reading list** rather than supplementary.

Plus the operations all have geometric meanings: addition is tip-to-tail, scaling is stretching (and reversing, for a negative scalar) — and every algebraic rule you will learn is a statement about those pictures, which is how to check whether a rule you half-remember is right.

2 Span

- **The span of a set of vectors is the set of all their linear combinations — every place you can reach by scaling them and adding — and it is always a line, a plane, a space, or (for the empty set) the origin.**
- **One non-zero vector spans a line through the origin; two independent vectors span a plane; three independent vectors in three dimensions span everything.**

- **And two vectors pointing along the same line span only that line, however you scale them — which is linear dependence, and is the failure this vocabulary exists to name.**
- **So: a set is linearly independent if none of its members lies in the span of the others — equivalently, if the only linear combination summing to the zero vector is the one with all coefficients zero, which is the form you actually compute with.**
- **Which matters in graphics immediately: three collinear points do not define a plane, and a mesh face built from them has no well-defined normal (Module 06 §2's zero-length cross product) — a degenerate triangle, which exists in every mesh you will ever load.**

3 Basis

A BASIS IS A SET THAT IS INDEPENDENT AND SPANS — SO EVERY VECTOR HAS EXACTLY ONE COORDINATE LIST

Spanning gives you at least one way to write every vector; independence guarantees at most one — and the two together give exactly one, which is what a coordinate system is and why both conditions appear in the definition.

Too few vectors and some points of the space are simply unreachable; too many and a vector has several different coordinate lists, so 'the coordinates of v ' stops naming anything — the basis is the condition that is just right.

And the dimension of a space is the number of vectors in any basis for it — which is the same number for every basis, and that invariance is a theorem rather than an observation, and a non-trivial one.

So 'three-dimensional' means 'every basis has three vectors' — not 'the vectors are written with three numbers', which is a fact about a coordinate convention rather than about the space. The distinction matters as soon as you meet a two-dimensional subspace of a three-dimensional space, which in graphics is every triangle.

4 In graphics

Concept	Where you will meet it	What goes wrong
Basis	Every coordinate frame: model, world, camera, tangent, light.	Mixing frames silently — see the note.
Span	The plane containing a triangle; the subspace a constrained joint can move in.	Degenerate faces that span only a line.
Linear independence	Whether three points actually determine a plane.	Zero-length normals, and the NaNs that follow when you normalise them.
Linear combination	Barycentric interpolation of colour, normals, and UVs across a triangle.	Weights that do not sum to one, which shifts everything subtly.
Dimension	Degrees of freedom in a rig, a spline, or a solver.	Over-constrained systems with no solution (Module 12).

Mixing frames silently is the characteristic bug of this whole subject — the numbers are valid, the arrows are in the wrong space, and nothing errors: you get a picture, and it is wrong in a way that looks like an art problem. Barycentric coordinates are the clearest everyday use of a linear combination, and they are the reason anything varies smoothly across a triangle.

AND THE HABIT THIS MODULE ASKS FOR

Label every vector with the space it lives in — in a comment, in a variable name, or at minimum in your head — because the type system will not do it for you and three floats look exactly like three other floats.

Which is the single cheapest defence against the characteristic bug of this subject (§4's note), and it costs a naming convention and nothing else.

And it makes Module 04 easy, because a change of basis is then visibly a conversion between two labelled spaces — `M_world_from_camera` — rather than a matrix appearing from nowhere with an unmemorable name.

So: `v_world`, `v_camera`, `n_tangent` — and never a bare `v`, which is a convention Project 2 enforces and which pays for itself the first time a shading bug turns out to be a frame mismatch.

Where to see this taught differently

Original material only goes so far — hearing the same idea in another voice is how it sticks. These are free, and they cover this module specifically.

Resource	What it covers for this module
3Blue1Brown, chapters 1 to 3 (free)	The whole module — vectors, span, and basis, visually, and this is the right first exposure.
MIT 18.06, lectures 1 and 5 (free video)	§§2 and 3 — the column picture and independence, from Strang.
Immersive Math, chapters 2 and 3 (free)	§§1 to 3 with manipulable figures — drag the basis vectors and watch the coordinates change, which is the point of §1.
Gortler, chapter 2 (library copy)	§4's frames — unusually careful about labelling which space a vector is in, which most graphics texts are not.

Exercises

Do these in code, not on paper, unless the exercise says otherwise. The module is not finished until they run.

- 1 **Draw ten vectors** and their sums, tip to tail.
- 2 **Write one arrow's coordinates** in two different bases.
- 3 **Find the span** of three given sets of vectors.
- 4 **Determine independence** for five sets, by solving for a zero combination.
- 5 **Construct three collinear points** and show they span only a line.
- 6 **Verify that a basis gives unique coordinates**, by finding them two ways.

- 7 **Find a set that spans but is dependent**, and one that is independent but does not span.
- 8 **Compute barycentric coordinates** of a point in a triangle.
- 9 **Interpolate a colour** across a triangle with them.
- 10 **Adopt the naming convention** and apply it to any existing code you have.

Self-check

Answer these in writing, from memory, without looking back at the notes. If you cannot, you have read this module but not yet learned it — that is useful information, not a failure.

- 1 What do a vector's coordinates mean?
- 2 Why does the same arrow have different coordinates in different bases?
- 3 Define span, and say what one, two, and three vectors can span.
- 4 Define linear independence, two equivalent ways.
- 5 Give the graphics consequence of dependence.
- 6 Define a basis and say what each condition buys.
- 7 What is dimension, and why is it well-defined?
- 8 What does 'three-dimensional' actually mean?
- 9 Name five places these concepts appear in graphics.
- 10 What is the characteristic bug, and what is the defence?